

Scansion type	Scansion	Comments
“Does the road wind uphill all the way?” (Christina Rossetti, Up-Hill 1)		
Classical scansion	wws/ss/ws/ws	2 substitutions for iambs: an anapaest (wws) and a spondee (ss). Four recognisable feet imply tetrameter, probably iambic.
Generative Metrics	wwss[sw]sws	The strong stress in the polysyllabic word ‘uphill’ is generally matched to a strong metrical stress. This, and the 9 syllable count, implies iambic tetrameter.
“May not the darkness hide it from my face?” (Christina Rossetti, Up-Hill 2)		
Classical scansion	ws/ws/ws/ws/ws	No substitutions – a regular iambic pentameter line
Generative Metrics	ws[sw]swwws	The strong stress in the polysyllabic word ‘darkness’, marked by brackets, is preceded and followed by weak stresses: a pattern which always matches the strong stress to a strong metrical stress. This, and the 10 syllable count, implies iambic pentameter.

Table 15: Generative Metrics and traditional scansions compared (Fabb, 2006)

“In metrical analyses such as Fussell (1979) as well as newer theories (e.g., Attridge 1982) the metrical form is molded to the rhythmic form, rather than being distinct from it...In the approach to metricality which sees rhythm as the expression of meter (Classical Scansion), these lines are well formed because the specific rhythms are permitted variations of iambic pentameter. In the approach to rhythm espoused in this paper (Generative Metrics), many aspects of these rhythms are simply irrelevant to the metricality of the lines; as we show below the lines are metrical because their numbers of syllables (nine and ten) can be counted by the metrical rules for iambic pentameter, and the polysyllables ‘uphill’ and ‘darkness’ are placed as they are in the line.”

One strand of Generative Metrics, which is led by the Slavic Metrists including most recently Tarlinskaja (1992, 1993, 1995 and 1997), and followed by investigators like Raabe (1975), Attridge (1982) and Duffell (2008), involves statistical analysis of stress patterns to determine which are more acceptable than others. For example, Tarlinskaja

(1993), marking weakest stress as 1, greatest stress as 5 and a caesura as ||, concludes that Shakespeare's earlier rhythm is typically [1 4 || 2 3 5], but his later rhythm is [1 4 3 || 2 5].

A second strand is primarily linguistic, and started with Halle and Keyser's proposal (1966 and 1971), revised by Levin (1973), that a fully stressed syllable between two unstressed syllables in the same word (termed a 'stress maximum') occurs only in positions where strong stress is expected in the meter. Magnuson and Ryder (1970 and 1971) and Kiparsky (1975) modified the theory to account for the fact that stressed monosyllabic words occur anywhere in the line not just in stressed positions, but polysyllabic words must place their stressed syllables in strong positions – that is, word boundaries are also a factor, not just lexical stress.

Table 16 shows the original Halle-Keyser predictions and some modifications made later – lines which are declared metrical should be frequently found, those which are unmetrical should be extremely rare.

Line/Stress/Meter		Halle-Keyser theory	Modifications
a	Line: Introduced grandfather to amuse friends Lexical: s w w \ s w w \ w \ w s \ s Metrical: W S I W S I W S I W S I W S (declared metrical although never found, but reclassified as unmetrical by the polysyllabic modification)	<u>metrical, but never found:</u> stress maximum ('-duced gra nd fath-') in WSW position	<u>unmetrical, therefore never found:</u> Strong stress in polysyllabic words not in S position ('in troduced ' = WSW, 'a muse' = SW)
b	Line: Ode to the West Wind by Percy Bysshe Shelley Lexical: s \ w \ w \ s \ s \ w \ s w \ s \ s w Metrical: W S I W S I W S I W S I W S W (declared unmetrical although found, and reclassified as metrical by Kiparsky)	<u>unmetrical, but found:</u> stress maximum in 'by Pe rcy' in WSW position	<u>metrical, and found:</u> monosyllables allowed to be stressed anywhere (stress maximum negated by stressed 'by' 'by Per cy' = ssw)
c	Line: Though death doth consume, yet virtue preserves Lexical: w \ s \ s \ w s \ s \ s w \ w s Metrical: W S I W S I W S I W S I W S (declared metrical, but which is unmetrical according to Kiparsky and others)	<u>metrical, but declared unmetrical by experts</u> (Groves, 2007b)	<u>unmetrical and found, but opposed:</u> 2 polysyllabic stresses in unstressed position (con sume =ws, lvirtue =sw), violating the polysyllabic modification

Table 16: Halle-Keyser theory and changes by Kiparsky, and Magnuson and Ryder

2.4.2 Stress Rules

This analysis was developed further by the discovery that stress in a word can be modified by the stress of closely associated contiguous words in the same phrase. This linguistic stress theory is of crucial importance to scansion, which requires a realistic assessment of the stress patterns in a verse. The idea was first proposed by Newman in 1946, according to Krifka (2001), but Chomsky and Halle (1968), followed by Liberman and Prince (discussed in Hayes, 1984) identified two processes for it – the Nuclear Stress Rule and the Compound Stress Rule (the role of an additional stress assignment rule, Stress Clash, is still being debated – see the Appendices, p194). The first rule proposes that words receive increasingly greater stress from the beginning to the end of a sentence in a recursive assignment of stress. This additional stress is

assigned to the principal stress within each stressed word – an illustration is given in the table below:

Stress Analysis	Stress Rule Applied
“Mary ate sweet ice cream”	
[Mary [ate [sweet [ice cream]]]]	Phrases
[Mary(1) [ate(1) [sweet(1) [ice(1) cream(1)]]]]	lexical stress
[Mary [ate [sweet [ice(1) cream(2)]]]]	compound stress
[Mary [ate [sweet(2) [ice(1) cream(3)]]]]	nuclear stress, first cycle
[Mary [ate(2) [sweet(3) [ice(1) cream(4)]]]]	nuclear stress, second cycle
[Mary(2) [ate(3) [sweet(4) [ice(1) cream(5)]]]]	nuclear stress, third cycle

Table 17: Stress Rules (Krifka, 2001)

In the sentence “Mary ate sweet ice cream” Nuclear Stress is assigned recursively from the last main stressed word back, so “Mary” receives a stress level of 2 and “cream” 5. “Ice” receives a stress level of 1 because it is the subordinate element of the compound “ice cream”; “cream” has the principal stress. According to Chomsky and Halle, the Compound Stress Rule assigns additional stress to the rightmost element in compounds of two or more nouns – for example, in “ice cream” principal stress is assigned to “cream”. Bolinger (1972), Sproat and Liberman (1987) and Giegerich (2006) modified it to accommodate some compounds, defined by their meaning, which are stressed on the leftmost element. For example, the first element of “metal bridge” is stressed because it expresses the substance of the second element, but “metal fatigue” is stressed on the second element because the relationship is one of association not substance: “fatigue in metal” not “fatigue made of metal”.

Beaver (1971) summarises the impact of these stress assignment rules on verse using stress-maxima to determine metricality:

“(1) The Compound Rule is apparently always operative (equivalently, never disregarded) in English verse.

(2) The Nuclear Stress Rule in its Noun phase apparently operates in... English, contrary to what some linguists have believed.”

2.4.3 Phonological Analysis

Another key component in understanding the nature of stress assignment and metrical patterns involved a new type of analysis partly dependent on syntax: phonological and clitic phrases. First applied to meter by Kiparsky (1975), the theory was defined by Selkirk in 1972 (see Selkirk, 1981, 1984 and 1986). Although researchers differ about how to divide some syntactic structures into phonological phrases, especially intonational units, (von Heusinger, 1999 and Atterer, 2000), the general principles are agreed. Table 18 gives phonological analyses by four researchers. However, I will use Selkirk's analysis because it is more accurate, easier to implement on computer, and is the only one used by experts in Phonological Metrics (Hayes, 1989 and Cureton, 1992). According to Selkirk, clitic phrases consist of a content word (a verb, adverb, adjective or noun) and any non-content words to its right and dependent upon it. Phonological phrases consist of clitic phrases which are syntactically related at the next level up. Intonational units consist of phonological phrases within breaks marked in the text such as commas, semi-colons or full stops (Hayes, 1989). Although Wheeldon (2000) notes that non-syntactic information can also influence the formation of phonological phrases (a phonological phrase will end immediately after an emphasised word), since this type of information cannot be referenced accurately by a computer I have ignored it.

Ref	Phonological Analysis
	clitic phrases are contained within square brackets [] phonological phrases are contained in curly brackets {} intonational units are contained in angle brackets <>
Abney	<{[Given] [the chance]}, {[rabbits] [reproduce] [quickly]}> <{[More than fifteen carpenters] [are working]}{[in the house]}>
Bachenko and Fitzpatrick	<{[Given] [the chance]}>, <{[rabbits]} {[reproduce] [quickly]}> <{[More than fifteen carpenters]}{[are working] [in the house]}>
Gee and Grosjean	<{[Given] [the chance]}>, <{[rabbits] [reproduce] [quickly]}> <{[More than fifteen carpenters]}{[are working]}{[in the house]}>
Selkirk/Cureton	<{[Given][the chance]}>, <{[rabbits]} {[reproduce] [quickly]}> <{[More than fifteen] [carpenters]} {[are working][in the house]}>

Table 18: phonological analyses of two sentences from Atterer (2000)

Selkirk's phonological theory accounts for changes in the way certain phrases are pronounced. For example "of" is never reduced to "o" when it occurs at the end of a clitic phrase, and stress clash is resolved within phonological units, but unresolved outside them (Hayes, 1989 and 1995, Kim, 1999 and Zuraw, 2006).

This analysis also predicts prosodic patterns. In 1971 the literary critic Hascall noted his impression that phonological and metrical boundaries coincide in iambic but not trochaic verse. Since then, more scientific analyses have been made: for example, Hayes (1989) has shown that the interaction of syntax and clitic phrases predict the metrical stress assignments in the phrases in Longfellow's *Hiawatha* (Table 19), and may have a more general application (Fitzgerald, 2007). Where the clitic phrase is separated by one syntactic break the clitic scansion is preferred, where there are three or more syntactic breaks, the non-clitic scansion is preferred, otherwise there is apparently free variation:

Clitic Phrases	Syntactic Breaks	Scansion preferred
[the great][lakes]	1: the _{NP} (great lakes)	- x – (clitic)
[in great][flocks]	2: In _{PP} (NP (great flocks))	- x – or x – x (either)
[that] [old feuds]	3: That _S (SUBJ(NP (old feuds))	x - x (non clitic)
NP= noun phrase; PP = prepositional phrase, S = sentence, SUBJ = subject		

Table 19: Hayes’s application of clitic phrases to scansion

This research has a direct impact on the identification of rhythm in poetry. According to Kiparsky (1975) phonological phrasing determines the location of caesurae in verse. More recent work on rhythm has accepted and generalised this conclusion: Barsch (1995) summarises:

“The view that the principles of versification are related to the prosodic features of a language is shared by so many metrists of totally different schools that it is a banality”.

These principles can identify rhythms in free verse - Gates and Laforge (1990) have successfully used them to identify T. S. Eliot’s “music of common speech”. They have also been used to determine the idiosyncratic rules of individual poets: Youmans (1983) and Harvey (1996) have developed these for Milton, Shakespeare and Donne.

2.5 Phonological Metrics and Recent Linguistic Theories

The principles also make useful observations about the relationship between metrical and lexical stress patterns. Literary scholars such as Andersen (1909) had noticed that stress patterns in words affect the rhythm of the line even when it is counter to the overall metrical rhythm, and Steele (1779) recorded the match between strong stresses in polysyllabic words and musical stresses. However these patterns were only defined scientifically relatively recently, by Magnuson and Ryder (1971) and Kiparsky (1975):

lexical stress matched metrical stress more frequently within polysyllabic words than monosyllabic ones. Kiparsky (1977) also found that stresses at the end of phrases were much more strictly matched than at the beginning of phrases. These observations support a basic principle of Generative Metrics formulated by Kiparsky in 1968 (“beginnings free, endings strict”), and have been confirmed by Hayes (1984a, 1996, 1998 and 2000), Tarlinskaja (1993 and 1997), Fabb (2001) and Li (2004).

To demonstrate the effectiveness of this principle, the table below (Table 20) gives the scansion of a line using Plamondon and Hartman’s systems, as well as a phonological analysis which results much more quickly in a firm identification of iambic pentameter.

Scansion		System
“My mistress, when she walks, treads on the ground” (Shakespeare Sonnet 130.12)		
a	My(1) mis(2) tress(0), when(2) she(1) walks(2) , treads(2) on(0) the(0) ground(2)	AnalysePoems
b	My(w) mis(s) tress(w), when(s) she(w) walks(s), ltreads(s) on(w) lthe(w) ground(s)	Scandroid
c	“[My(w) mis(s) tress(w)], [when(w) she(w) walks(s)], l[treads(s)] [on(w) lthe(w) ground(s)]”	Cureton’s phonological analysis

Table 20: scansion of a line by AnalysePoems, Scandroid and Cureton

Whereas Scandroid and AnalysePoems need to reference the stress patterns in each of the five feet to determine the overall metrical pattern, the phonological scansion procedure only references stresses at the ends of the phonological units. The closing syllables of three of the four phonological units fit with the expected metrical stress patterns of iambic pentameter: “mis(s)|tress(w)”, “|she(w) walks(s)|” and “lthe(w) ground(s)|”. Phonological theory claims that these are the key areas the listener targets to identify metrical patterns (Hanson, 1991, Hayes, 1989 and followed by Dahlgren, 2005). Hayes and Kaun (1996) conducted experiments on song lyrics leading to two conclusions which help in identifying poetic meter.

1. metrical and lexical stresses almost always coincide at the end of phonological phrases. In fact, the higher up the hierarchy, the more likely the match: clitic phrases 90% of the time, phonological phrases 97% and intonational units 99%. See Table 21b for an example. These conclusions are also backed up by recent research into matching lyrics to music, termed “textsetting”: Halle (2004) notes that words grouped together in musical phrases sound most natural to audiences when the words also form phonological units – musical groupings rarely cross phonological boundaries.

Ref	Scansion	System
“Or how haps it I seek not to advance” (Shakespeare Henry VI Part 1 3.1.31)		
a	Or(w) how(s)/ haps(s) it(w)/ I(w) seek(s)/ not(w) to(s)/ ad(w)vance(s)	Word stress
comment	(1 foot is a trochee, 4 feet are iambs: ws/sw/ws/ws/ws)	
b	<[Or(w) how(s)] _{CP} / [haps(s) it(w)] _{CP} } _{PP} / {[I(w) seek(s)] _{CP} [not(w) to(s)/ ad(w)vance(s)] _{CP} } _{PP} > _{IU}	Prosodic stress
comment	Using the last two stresses in each phonological unit gives ws]/sw]}/ws]/xx/ws]}}> (where x is a stress of irrelevant value). 3 clitic phrases match metrical stresses (ws/), as well as one clitic phrase’s associated phonological phrase and intonational unit. Only the second clitic phrase (and its associated phonological phrase) does not match the iambic metrical stress: [haps(s) it(w)]}	
clitic phrases are labelled ‘CP’ and contained in square brackets []; phonological phrases are labelled ‘PP’ and contained in curly brackets {}; intonational units are labelled ‘IU’ and contained in angle brackets <>		

Table 21: word and prosodic stress (from Hayes and Kaun, 1996)

2. Hayes and Kaun (1996) also note (following Kiparsky, 1975) that metrical and lexical stresses coincide most often within polysyllabic words (98% of the time there is a match): for example in Table 21a, whereas the lexical stress in disyllabic words matches a iambic metrical scheme (‘ad(w)vance(s)’), in monosyllabic words it does not necessarily match (‘haps(s) it(w)’).